This workshop proposal was proposed to IEEE VR 2027.

First-Person eXeRtion (FPXR): XR Visualization and Interaction in Motion

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1 Workshop Description

We propose First-Person eXeRtion (FPXR), a half-day workshop exploring how XR visualizations can support people in understanding and acting on data while participating in physical activities, such as sport and exercise. Here, first-person refers to visualizations intended for use by the physically active participants themselves, rather than by spectators or external analysts. The workshop centres on how bodily movement, physical exertion, and attention to the ongoing activity shape what information people need, how it should be visualized, and how they can interact with it. Relevant tasks may include monitoring effort, comparing current performance with a target, and interpreting information about the surrounding environment to inform immediate decisions.

Bringing together researchers and practitioners in XR, visualization, HCI, and sports science, FPXR aims to: (a) characterize spatial design considerations, including layout, positioning, and anchoring, for visualizing different data types in XR during physical activity; (b) envision XR visualizations and corresponding interaction techniques that support these tasks while accommodating movement, exertion, and competing attentional demands; (c) propose evaluation methods that examine both how effectively these designs support data understanding and interaction, and how they affect the ongoing physical activity, including participants' performance, experience, and safety.

2 Motivation

Sports and physical activity have attracted interest across visualization, HCI, and XR. Visualization workshops, such as the first workshop on sports data visualization at IEEE VIS 2013, emphasized data exploration for fans, media, and team officials. HCI workshops have approached this area through bodily awareness and motivation, such as the workshop on HCI and Sports at CHI 2014 [12]; embodied interaction, enjoyment, and flow, such as the workshop on Embodied Interaction in Sports at CHI 2021 [16]; and AR interaction in dynamic physical and social contexts, such as the AR On-the-Move workshop at CHI 2026. These initiatives provide complementary foundations in sports data analysis, embodied interaction, and mobile XR experiences.

FPXR connects these visualization and HCI initiatives, considering interaction as an integral part of visualization. Its closest predecessor is the workshop on First-Person Visualizations for Outdoor Physical Activities [14] at VIS 2024, which centered around real-time, first-person visualization and interaction. Its subsequent agenda highlights how environments, social situations, and available input/output modalities shape data use [13]. FPXR builds on this agenda through a focused examination of XR visual representations, active data interaction, and evaluation under the combined demands of movement and exertion.

The workshop includes everyday movement and exertion beyond sport. It relates to the broader research agenda of *Visualization in Motion* [17], which discusses and categorizes the relative movement relationships between a viewer and a visualization. FPXR draws on this agenda while focusing specifically on situations in which the person using the visualization is physically engaged in an ongoing activity. Further motivation comes from *SportsXR* [11], which explores immersive analytics for situational awareness and decision-making in sports, and *Running with Data* [3,4], which combines a survey with runner-centered workshops to explore the design space of in-situ running visualizations and interactions. Building on these initiatives, FPXR focuses on real-time, first-person XR visualization and interaction in movement and exertion scenarios.

3 Themes and Topics of Interests

Our workshop interests include the following three themes:

Designing situated XR visualizations in motion. This theme concerns how different data types are visually represented, spatially arranged, and anchored in XR during physical activity. Existing studies have explored situated visual feedback for basketball shooting [10], embedding visualizations in swimming broadcasts [15, 19, 20], and in running for interval training [5]. Building on these examples, we consider how spatial design can accommodate data and moving viewers. How should different data types, for example, static data & dynamically updating data, ordinary data & continuous data, be represented and combined within a coherent spatial layout? How should visualizations be positioned relative to the participant's head, body, or surrounding environment? How should viewing distance, relative position between the visualization and the viewer, as well as their world coordinates, be configured to support legibility while limiting visual clutter and occlusion of activity-relevant surroundings? How can layouts preserve readability across multiple visualizations as the participant's position and viewpoint change, and when should these layouts remain stable or adapt to changing viewing conditions?

Interacting with XR visualizations beyond monitoring. This theme concerns input techniques that allow participants to intentionally access and manipulate data during physical activity. Recent work on immersive running visualizations [6] motivates interactions

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beyond monitoring, such as querying and filtering data, comparing observations, annotating events, and reconfiguring views. Research on direct-touch selection while running also highlights limitations of conventional tapping interactions under motion [3]. Another work in progress on information enhancement for squash players [21] augmented squash practice with ball-, rule-, and event-related data across modalities, while in the evaluation, their participants showed a need to manipulate augmented visual and auditory cues, such as temporally turning off some of them. This body of work raises a set of interesting questions, such as: which XR input modalities, or combinations of modalities, are suitable for these operations? How can interaction techniques fit into an ongoing activity while minimizing interruption and additional physical effort? How can systems distinguish intentional commands from movements inherent to the activity, and allow participants to confirm or correct their actions?

Evaluating XR visualizations and interactions in motion. This theme concerns the criteria and methods for assessing visualization and interaction techniques during physical activity. Studies of immersive running visualizations and broader visualization-in-motion experiences motivate considering task performance alongside user experience [5, 18]. Which measures can capture data understanding and interaction reliability, as well as effects on the ongoing activity, including performance, workload, experience, and safety? How should studies balance experimental control with realistic movement, exertion, and environmental conditions? How can logging, sensing, observation, and appropriately timed self-reports capture these outcomes without substantially altering the activity being studied?

4 Submission Format and Review

Submissions should identify a question relevant to any of the 3 themes listed in Sect. 3. We welcome original research, including but not limited to work in progress, design studies, and position papers. Authors are encouraged to prepare their submissions using the official template. The recommended submission length is between 1000 and 2500 words, but longer submissions will be considered. Accepted papers are to be archived in the IEEE Digital Library. Submissions will be collected through an alternative online platform, such as Google Forms, rather than through PCS.

Workshop submissions will be reviewed internally by the workshop organizers. We do not plan on having a pool of reviewers or a program committee beyond the organizers, given that the review process will be lightweight: (1) All workshop organizers read all submissions that they are not in conflict with, (2) All non-conflicted organizers indicate if a submission is a) of the requested submission types, b) on topic and likely to stimulate interest and discussion, c) is of reasonable presentation quality. (3) All submissions for which a majority of non-conflicted reviewers answer 'yes' to these three questions will be accepted. We will follow up with rejected submissions to offer the authors an opportunity to revise and resubmit - something that is made feasible thanks to our flexible timeline and submission format. In case a conflict cannot be resolved internally, we will seek external reviewers to handle these cases.

5 Tentative Schedule and Participants

Participants & Audiences: We expect 25-30 participants, with an optimal number of 7-15 accepted submissions. We will advertise this workshop through the mailing lists, Slack groups, and social media channels of IEEE VR/ISMAR/VIS and CHI communities and our personal networks. We will encourage contributions from PhD students who work on relevant topics, practitioners from visualization, XR, and sports science, and people whose lived experience of physical activity can inform design.

Participation and intended outcomes: All attendees will contribute to challenge mapping, group discussion, and plenary synthesis. We will use structured turn-taking, accessible written prompts, and opportunities to contribute through either speech or writing. The organizers will also host an online session simultaneously for participants that are unable to attend in person. The primary intended outcome of this workshop is to establish an initial list of challenges and opportunities and to identify a group of authors that will work together on a joint publication that lays out a research agenda.

Timeline & Activities: The workshop will run for three hours. The indicative morning schedule below can be shifted to an afternoon slot. No keynote is planned as we aim to prioritize hands-on activities, active participation, and collaborative discussion.

- 09:00–09:15 **Opening and orientation:** introduce the scope, intended outcomes, and themes; record participant interests on a shared board, which will be used to collect questions and document discussions throughout the workshop.
- 09:15–10:15 **Short presentations and questions:** each presenter introduces their submission and concludes with an open challenge for discussion, followed by a brief Q&A. Other participants record issues on the shared board.
- 10:15–10:45 **Challenge mapping:** participants add questions and challenges based on the presentations and personal experiences, which will be clustered into three topics. The workshop themes provide starting points as participants refine each discussion question. Participants will then be grouped based on the topic they are interested in.
- 10:45–11:00 **Coffee break.**
- 11:00–11:30 **Small-group discussion:** each group examines its question through a concrete scenario of its choice. An organizer will facilitate the discussion, and a Zoom session with breakout rooms for online participants. Each group nominates a rapporteur to record in a shared template the scenario, movement-related challenges, and two or three research questions or possible approaches.
- 11:30–11:45 **Group reports:** each group will have up to five minutes to present its discussion question, illustrate the challenge through its chosen scenario, and summarize its proposed research directions. Group notes will be added to the shared board so that all participants can review the outcomes.
- 11:45–12:00 **Plenary synthesis and closing.** participants compare the group outcomes, identify connections and differences across the topics, and agree on a small set of priorities for future research. The organizers will summarize these priorities and share the workshop notes.
- 12:00 onwards **Optional workshop lunch.** Interested participants may continue discussions and networking over lunch.

6 Workshop Organization and Timeline

The timeline for the workshop organization is as follows. All deadlines are at 11:59 pm (23:59) AoE time — October 20, 2026: **Call for Participation**; December 20, 2026: **Submission Deadline**; January 3, 2027: **Internal Reviews Completed**; January 7, 2027: **Author Notification**; January 12, 2027: **Camera Ready Version**.

7 Equipment and Schedule

We request a presentation room with a microphone, a projector, tables, and movable chairs, and power outlets. We will also need pens and papers. We propose an in-person workshop, but allow online participation. The primary organizer will coordinate additional workshop commitments with the chairs and report changes.

8 Organizer's Background, Related Publications and Research

Ang Li, angli21@sigchi.org | (`http://angli.phd/`): Ang 'Leon' Li is a postdoctoral researcher at Inria, France. His research interests revolve around the field of HCI and Data Visualisation with a focus

on Immersive Analytics. His work examines situated visualisation for intensive movement scenarios such as running [4–6], and spatial interaction under movement [3].

Lijie Yao, yaolijie0219@gmail.com | (https://lijieyao.com): Lijie Yao is an Assistant Professor at Xi'an Jiaotong-Liverpool University. Her work focuses on exploring the impact of motion and spatial factors on visualization perception, design, prototyping, and interaction. She proposed *visualization in motion* and demonstrated that simple visual representations could be read even under high speeds and irregular trajectories [17], explored how to embed and design *visualizations in motion* in real application scenarios like swimming [15, 19, 20], video games [18].She is the PC for VIS from 2024 and co-organized the FPVis workshop.

Yi Li, yi.li@cs.au.dk: Yi is a postdoc researcher at Aarhus University, Denmark. Her research interests revolve around the field of Computational Interaction in XR. Her work develops ergonomic user interfaces adaptive to human physical capability, such as muscle fatigue and visual strains.

Linjia He, linjia.he@monash.edu: Linjia He is a PhD student at the Exertion Games Lab, Monash University, Australia. Her research interests lie in augmented reality and its applications in user experience, especially for on the go. Her PhD project investigates how AR can be applied to cycling, with a particular focus on its interface design and interaction [2]. She co-organized theAR On-the-Move workshop at CHI'26.

Tica Lin, mlin@g.harvard.edu | (https://ticalin.com): Tica Lin is a Senior Research Scientist at Dolby Laboratories. Her research interests include immersive analytics, human-AI interaction, and SportsXR [7, 11]. Her work addresses challenges in presenting data in dynamic physical environments for situated analytics. She has investigated the design of XR visualizations in the physical contexts for sports applications, including first-person visualizations for motor skill training [10] and interactive spatiotemporal data visualizations for live games [1, 8] , among other real-world contexts [9]. Her research closely aligns with the aims of this workshop.

Yalong Yang, yalong.yang@gatech.edu | (https://ivi.cc.gatech.edu/): Yalong Yang is an Assistant Professor in the School of Interactive Computing at Georgia Tech. His interests encompass fundamental visualization and HCI challenges in VR/AR, as well as applied domains such as sports. He has extensive experience in building sportsXR applications [7, 8, 10, 11].

Charles Perin, cperin@uvic.ca | (http://charlesperin.net/): Charles Perin is an Associate Professor of Computer Science at the University of Victoria. His research focuses on designing and studying new interactions for visualizations and on understanding how people may make use of and interact with visualizations in their everyday lives, including mobile and physical visualization. Relevant to this workshop, he co-led the FPVis workshop [14] and the resulting publication [13]. He has published several papers on the topic of sports visualization in the past decade and is conducting research on beyond-desktop visualizations [4–6]

Maxime Cordeil, m.cordeil@uq.edu.au |(https://sites.google.com/view/cordeil/): Maxime Cordeil is an associate professor in XR, HCI and Data Visualisation at the University of Queensland, Australia. His research explores how VR/AR technologies enable users to better understand and interact with complex data. Relevant to this workshop, he has extensive experience in Immersive Analytics for movement scenarios [3–6].

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